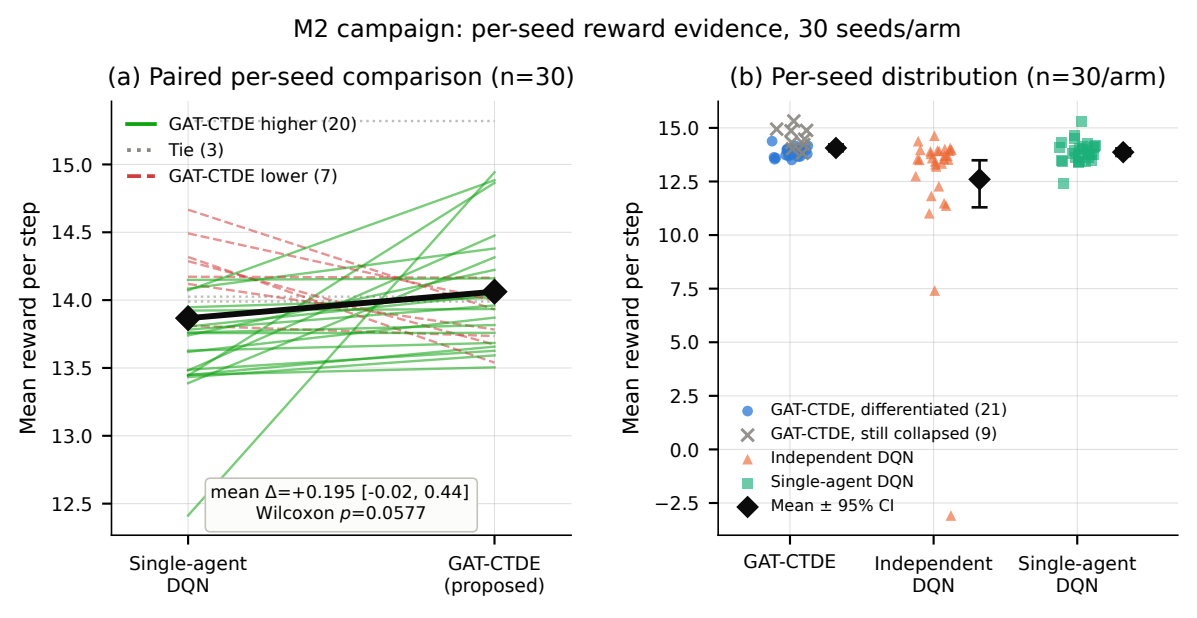

M2 campaign: per-seed reward evidence, 30 seeds/arm
(a) Paired per-seed comparison (n=30)
GAT-CTDE higher (20)
Tie (3)
GAT-CTDE lower (7)
Mean reward per step
15.0
14.5
14.0
13.5
13.0
12.5
mean Δ=+0.195 [-0.02, 0.44]
Wilcoxon p=0.0577
Single-agent DQN
GAT-CTDE (proposed)
(b) Per-seed distribution (n=30/arm)
Mean reward per step
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
GAT-CTDE, differentiated (21)
GAT-CTDE, still collapsed (9)
Independent DQN
Single-agent DQN
Mean ± 95% CI
GAT-CTDE
Independent DQN
Single-agent DQN